

3D-bioprinted functional and biomimetic hydrogel scaffolds incorporated with nanosilicates to promote bone healing in rat calvarial defect model

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ABSTRACT

Three-dimensional (3D) bioprinting is an extremely convenient biofabrication technique for creating biomimetic tissue-engineered bone constructs and has promising applications in regenerative medicine. However, existing bioinks have shown low mechanical strength, poor osteoinductive ability, and lacking a suitable microenvironment for laden cells. Nanosilicate (nSi) has shown to be a promising biomaterial, due to its unique properties such as excellent biocompatibility, degrade into nontoxic products, and with osteoinductive properties, which has been used in bone bioprinting. However, the long term bone healing effects and associating risks, if any, of using nSi in tissue engineering bone scaffolds *in vivo* are unclear and require a more thorough assessment prior to practical use. Hence, a functional and biomimetic nanocomposite bioink composed of rat bone marrow mesenchymal stem cells (rBMSCs), nSi, gelatin and alginate for the 3D bioprinting of tissue-engineered bone constructs is firstly demonstrated, mimicking the structure of extracellular matrix, to create a conducive microenvironment for encapsulated cells. It is shown that the addition of nSi significantly increases the printability and mechanical strength of fabricated human-scale tissue or organ structures (up to 15 mm height) and induces osteogenic differentiation of the encapsulated rBMSCs in the absence of *in vitro* osteoinductive factors. A systematic *in vivo* research of the biomimetic nanocomposite bioink scaffolds is further demonstrated in a rat critical-size (8 mm) bone defect-repair model. The *in vivo* results demonstrate that the 3D bioprinted nanocomposite scaffolds can significantly promote the bone healing of the rat calvarial defects compared to other scaffolds without nSi or cells, and show rarely side effects on the recipients. Given the above advantageous properties, the 3D bioprinted nanocomposite scaffolds can greatly accelerate the bone healing in critical bone defects, thus providing a clinical potential candidate for orthopedic applications.

1. Introduction

Three-dimensional (3D) bioprinting, also called 3D cell printing [1], is an innovative biotechnology for the rapid biofabrication of biomimetic constructs using cell-laden hydrogels (also called “bioink”) that can be cross-linked to form a hydrated matrix microenvironment for encapsulated cells [1,2]. Compared with traditional 3D printed cell seeded scaffolds, the greatest advantage of 3D bioprinting is their ability to deliver multiple types of cells *via* an accurate, controlled

approach to 3D engineered constructs [2,3]. Among the three primary recently developed bioprinting technologies (inkjet, extrusion and laser-assisted bioprinting), the most common and affordable technology is extrusion bioprinting, and at the same time, it shows the advantage to deposit bioinks with a very high cell density [2]. However, the development of extrusion-based 3D bioprinting has reached a bottleneck due to the lack of appropriate bioinks with high mechanical strength, printability, and biocompatibility [3,4]. The ultimate aim of bioprinting is to develop human-scale functional tissues or organs with high

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fidelity, which requires excellent printability along the Z-axis, for use as *in vivo* transplants to alleviate organ shortages. However, recently developed bioinks, such as collagen [5], gelatin [6], alginate [6,7], hyaluronic acid [8], and decellularized extracellular matrix [9], enable the precise deposition of viable cells but are hindered by limited mechanical strength and structural integrity for supporting a greater number of layers.

Gelatin (Gel), the product of collagen degradation, is a low-cost biomaterial with good biocompatibility. As gelatin retains the Arg-Gly-Asp (RGD) sequence from collagen, it can promote cell proliferation, migration [10]. Although gelatin can be gelled by thermally induced crosslinking, it can easily dissolve at 37 °C, and therefore, gelatin is rarely used for bioprinting in its native form [11]. To avoid dissolution and to increase its mechanical strength, gelatin is often blended with other biomaterials [6,9], or conjugated with methacrylate groups to its amine side groups through a grafting reaction to obtain photo-cross-linked gelatin methacrylate (GelMA) [12,13]. Alginate (Alg), an anionic polysaccharide polymer, is a popular biomaterial used in bioprinting due to its low cost, biocompatibility, and convenient cross-linking in calcium chloride (CaCl₂) solutions [6,14]. However, alginate is a bioinert material, as mammalian cells do not produce specific enzymes to degrade this polymer; thus, it can be retained *in vivo* for a long time [7]. In addition, due to the lack of RGD sequences, cell attachment is limited. Therefore, RGD peptide modification [15] is often performed to obtain desirable cellular functions, although this procedure increases the complexity and cost, which limits the applications of alginate.

In recent years, osteogenesis or angiogenesis induced by elements or ions has received increased attention. The promising effects of certain elements or ions decrease the cost of osteogenic or angiogenic induction in cultures *in vitro*, which may accelerate their clinical transformation for biofabrication. Lithium has been demonstrated to activate β -catenin-mediated T cell factor (TCF)-dependent transcription during bone and cartilage fracture healing [16]. Magnesium can induce nitric oxide production in endothelial cells, which is essentially the same mechanism by which vascular endothelial growth factor (VEGF) induces angiogenesis [17]. Zhang et al. reported that a nanocomposite hydrogel stabilized by self-assembled multivalent bisphosphonate-magnesium nanoparticles mediated the sustained release of Mg²⁺ and promoted *in situ* bone regeneration [18]. Moreover, silicon has been shown to induce angiogenesis by upregulating nitric oxide synthase (NOS), leading to increased VEGF production [19]. Orthosilicic acid has been demonstrated to promote collagen type I synthesis and osteoblast differentiation *via* the Wnt/ β -catenin signaling pathway, and the role of Wnt signaling has been associated with osteogenic differentiation of stem cells [20]. Hence, the use of beneficial elements to enhance osteogenic induction is a promising trend for decreasing the cost of osteoinductive factors.

Laponite XLG nanosilicate (nSi, Na_{0.7}[(Mg_{5.5}Li_{0.3})Si₈O₂₀(OH)₄]_{0.7}) is a two-dimensional (2D) synthetic nanomaterial composed of disk-shaped nanoscale crystals with a high aspect ratio (25–30 nm in diameter and approximately 1 nm in thickness). The surfaces of the crystals are negatively charged, and the edges are positively charged [21]. These features make nSi able to strongly interact with anionic, cationic, and neutral polymers to form a physically cross-linked network with shear-thinning characteristics that are critical for printability during the extrusion bioprinting process and, thus, significantly improve the mechanical strength of printed products [21,22]. Therefore, strong physical interactions between nSi and polyampholytic gelatin were expected, and injectable nanoclay-gelatin hydrogels were developed in physiological media [22]. Additionally, nSi can also be used as a rheology modifier of alginate solutions to increase its printability [23]. A bioink with strong physical interactions composed of a cationic-modified nSi and an anionic polymer alginate and gellan gum was previously reported to exhibit significantly improved mechanical properties, printability, and printing fidelity [24]. Furthermore,

Laponite is an FDA-approved material that is generally regarded as safe, and its degradative bioactive products, such as Mg²⁺, orthosilicic acid, and Li⁺, have been reported to promote osteogenic differentiation of human mesenchymal stem cells (hMSCs) in 2D culture [25]. In addition, when adding nSi to GelMA solution, the generated hybrid hydrogels were capable of promoting osteogenic differentiation of seeded MC3T3-E1 cells without the addition of growth factors [26]. Even though previous studies have demonstrated that nSi is biocompatible, making it suitable for cell culture, bone tissue engineering, and other biomedical applications. However, the long term bone healing effects and associating risks, if any, of using nSi in 3D bioprinted tissue engineering bone scaffolds *in vivo* are unclear and require a more thorough assessment prior to practical use.

Hence, in this study, we firstly developed a gelatin-alginate-based functional and biomimetic Laponite nanocomposite bioink with high printability and mechanical strength for 3D bioprinting to biofabricate tissue-engineered bone constructs in the absence of osteoinductive factors. We also take advantage of the bioinert property of alginate to prolong the retention of bioprinted scaffolds *in vivo* to avoid rapid degradation by proteases [7] to ensure the degradation rate of the scaffold well matches the rate of new bone formation and the systematic *in vivo* results demonstrated the effective bone repair of the 3D bioprinted nanocomposite scaffolds compared to other scaffolds without nSi or cells, and show rarely side effects on the recipients. As the electrostatic interaction of nSi with gelatin and alginate enhanced the mechanical strength of the bioink [21–23], the 3D bioprinted tissue-engineered bone constructs could be printed up to more than 1 cm (up to 60 layers and 15 mm height) along the Z-axis, which is highly promising for the future development of bone tissue-engineered human-scale functional tissues or organs for translation to the clinic.

2. Materials and methods

2.1. Hydrogel preparation, physicochemical characterization and scaffold fabrication

2.1.1. Materials

Sodium alginate (A0682-100G), gelatin (48722-500G-F), calcium chloride (793639-100G), and sodium citrate (PHR1416-1G) were obtained from Sigma-Aldrich (St. Louis, MO, USA). Nanosilicate (Laponite XLG) was obtained from BYK-Chemie GmbH (Wesel, Germany).

2.1.2. Hydrogel preparation

Gelatin (Gel) granules, sodium alginate (Alg) and Laponite nanosilicate (nSi) powders were sterilized by ethylene oxide. nSi was slowly added to sterilized deionized water and mixed homogeneously using a sterilized magnetic stir rod for 60 min at room temperature to prepare a 20 mg/ml, 40 mg/ml or 60 mg/ml nSi solution, respectively. A blended solution composed of 200 mg/ml Gel and 20 mg/ml Alg was prepared in deionized water at 40 °C. The different concentration nanocomposite hydrogel was obtained by 1:1 mixing of 20% w/v Gel + 2% w/v Alg solution with the 2% w/v, 4% w/v or 6% w/v nSi to obtain a final concentration of 10% w/v Gel + 1% w/v Alg + 1%/2%/3%w/v nSi solution (referred to as 1%nSi, 2%nSi and 3%nSi, respectively). The 10% w/v Gel + 1% w/v Alg solution (referred to as 0%nSi) was obtained by 1:1 mixing of 20% w/v Gel + 2% w/v Alg solution with sterilized deionized water (Table 1).

2.1.3. Cryo-TEM

In order to take the cryo-TEM (FEI Talos F200C, America) images of nSi, 2% w/v nSi suspensions were prepared with deionized water, then the suspensions were mounted on a holey carbon copper grid having 200 mesh size. Subsequently, the grid was frozen in liquid ethane and transferred to cryoholder for imaging.

Table 1
The final concentration of the nanocomposite hydrogels.

	Gelatin (w/v%)	Alginate (w/v%)	Nanosilicate (w/v%)
0%nSi	10	1	0
1%nSi	10	1	1
2%nSi	10	1	2
3%nSi	10	1	3

2.1.4. Fourier transform infrared spectroscopy (FT-IR)

FT-IR of the two lyophilized hydrogel samples (0% nSi and 2% nSi) and pure nSi powder were performed using a VERTEX 33 Bruker FT-IR system (Germany).

2.1.5. Hydration degree test

To evaluate the hydration degree of the hydrogels, samples ($n = 6$) were prepared and immersed in phosphate-buffered saline (PBS) for 24 h, and the wet weight (M_o) was measured. All samples were then lyophilized to obtain the dried weight (M_i). The hydration degrees of the hydrogels were evaluated using the following equation:

$$\text{Swelling ratio} = (M_o - M_i)/M_i \times 100\%.$$

2.1.6. Microstructure and element analysis

The morphology and elemental distribution of the hydrogels were observed via a scanning electron microscope (SEM) (S-3400; Hitachi, Tokyo, Japan) equipped with a Bruker XFlash 5030 energy-dispersive X-ray (EDX) spectrometer. Hydrogel samples ($n = 6$) were frozen at -80°C , cracked with a razor blade, and lyophilized. EDX mapping was performed at an accelerator voltage of 20 keV. Next, all samples were fixed and sputter-coated with gold to a thickness of 21 nm, and SEM was performed under identical conditions at an accelerator voltage of 5 keV.

2.1.7. Rheological analysis

Rheological tests were performed using a Thermo Scientific HAAKE rheometer (Thermo scientific, MARS 40/60, Germany) with a cone-plate geometry measuring system (25 mm diameter, 2° cone angle, 99 truncation gap). Flow rate sweeps were performed at a shear rate from 10^{-2} to 10^2 s^{-1} . To measure the yield stress, oscillation stress sweeps were performed at 1 Hz from 10^{-1} to 10^3 Pa . The shear rate sweeps were performed at 37°C and the shear stress sweeps were performed at both 37°C and 25°C .

2.1.8. Mechanical test

To determine the modulus of the hydrogels, uniaxial compression testing was performed with a model 858 material testing system (MTS Systems, Minneapolis, MN). All tests were implemented at 40% compression and a strain rate of 10 mm/min. The compressive modulus was calculated from the slope in the toe region corresponding to 0.05–0.10 strain. Samples ($n = 6$) were prepared by pouring liquid hydrogels (at 37°C) into a pre-3D-printed hollow cylinder (10 mm inner diameter, 5 mm height), cooling to room temperature, and then placing into a 2% w/v calcium chloride (CaCl_2) solution for 10 min. Samples were then transferred to a $1 \times$ PBS solution, washed three times, and then immersed for 24 h. All samples were tested at room temperature.

2.1.9. 3D printing process

A Particle Cloud (PC) Bioprinter (PC Printer; Particle Cloud Biotechnology, Xi'an, China) was used for printing all constructs. The nanocomposite hydrogel (2%nSi) was loaded into a VOL-30 extruder cartridge (PC Printer; Particle Cloud Biotechnology, Xi'an, China) wrapped with a heating unit, and the temperature was set at 37°C . A 22-gauge blunt-tipped stainless-steel needle was equipped. STL files were generated from 3D CAD drawings created in SolidWorks and converted to G-code using Slic3r. G-code files were then input into the

PC Bioprinter software (PCLab), which controlled the printer and the print parameters. Unless otherwise stated, the printing process was performed under the following parameters: a print head speed of 50 mm/s, a nozzle height of 300 μm from the print surface, and a flow rate of 12.72 ml/h. Constructs were printed onto homothermal metal plates (25°C). After printing, the constructs were cross-linked with 2% w/v CaCl_2 solution for 10 min and then transferred to $1 \times$ PBS and rinsed three times.

2.1.10. Scanning electron microscopy (SEM)

The structural analysis of 3D printed pure scaffolds (PPS) was performed using a SEM (S-3400; Hitachi, Tokyo, Japan). Samples ($n = 6$) were dehydrated through an ascending ethanol series (diluted with deionized water) ending in 100% ethanol and dried at room temperature in a vacuum. All samples were fixed and sputter-coated with gold for SEM.

2.2. In vitro cell culture and cytocompatibility investigation

2.2.1. Cell isolation and incubation conditions

Bone marrow mesenchymal stem cells (BMSCs) were isolated from the femoral shaft of 2-week-old Sprague Dawley (SD) male rats and expanded to passage 3 in α -minimum Eagle's medium (α -MEM) supplemented with 10% v/v fetal bovine serum (FBS), 100 U/ml penicillin, and 100 mg/ml streptomycin (all from Gibco Biosciences, Dublin, Ireland) at 37°C with 5% CO_2 . When approximately 80%–90% colony formation was observed, BMSCs were trypsinized, counted, and collected for the subsequent *in vitro* and *in vivo* experiments.

2.2.2. 3D bioprinting

For 3D bioprinting, approximately 3×10^7 rBMSCs were harvested, suspended in 600 μl cell culture medium and uniformly mixed into prepared 10 ml 0%nSi or 2%nSi hydrogel to obtain a final cell concentration of $3 \times 10^6/\text{ml}$. 3D bioprinted rBMSCs-laden scaffolds (15 mm $\times$ 15 mm $\times$ 1.5 mm, 6 layers) were biofabricated with the same parameters described above, cross-linked in 2% w/v CaCl_2 solution for 10 min, and then washed with $1 \times$ PBS three times. The scaffolds were then placed in normal medium (NM, α -MEM supplemented with 10% v/v FBS, 100 U/ml penicillin, and 100 mg/ml streptomycin) or osteoinductive medium (OIM, NM supplemented with 10 mM β -glycerophosphate, 2 mM glutamine, 0.2 mM ascorbic acid, and 100 nM dexamethasone), and incubated at 37°C with 5% CO_2 . The medium was changed every two days.

2.2.3. Live/dead assay

After culturing *in vitro* for 1, 3, 5, 7, 14 and 21 days, cell viability was assessed using a LIVE/DEAD™ viability/cytotoxicity assay kit (Invitrogen, Bioscience, Ireland). Briefly, the bioprinted scaffolds (2% nSi) ($n = 3$) were washed in PBS three times and then incubated in α -MEM containing 2 mM calcein AM (green fluorescence for live cells) and 4 mM ethidium homodimer-1 (red fluorescence for dead cells). Then, scaffolds were washed again with PBS three times, and images were captured via a laser scanning confocal microscope (LSCM) (Nikon A1R, Japan).

2.2.4. Cell cytoskeleton observation

To detect actin filaments, 3D bioprinted scaffolds (2%nSi) ($n = 3$) were cultured in NM for 7 days, then washed three times in PBS. Subsequently, samples were fixed with 4% paraformaldehyde for 20 min and permeated with 0.5% Triton X-100 (Sigma Aldrich, USA) for 5 min. After rinsing in PBS three times, the scaffolds were stained with rhodamine/phalloidin (Dojindo, Kumamoto, Japan) at room temperature and protected from light for 30 min. DAPI (Dojindo, Kumamoto, Japan) was used for double-staining of the cell nuclei, and images were captured using a LSCM (Nikon A1R, Japan).

2.2.5. Hematoxylin and eosin (H&E) staining

H&E staining was carried out for sections of the 3D bioprinted scaffolds (2%*n*Si). After a series of standard procedures for embedding the sections in paraffin, the paraffin-embedded sections were deparaffinized by toluene, rehydrated through a graded series of ethanol and rinsed in deionized water. Subsequently, H&E staining was performed, and sections were observed using a microscope imaging system (BX53; Olympus, Japan).

2.2.6. Cell proliferation assay

Cell proliferation was investigated using a CCK-8 assay (Dojindo, Kumamoto, Japan). A volume of 1 ml of 10% CCK-8 solution in NM was added to each sample ($n = 3$) at a specified time point, and the samples were incubated for 4 h. After that, 300 μ l of the culture medium from each sample was transferred to a 96-well plate; each well had a volume of 100 μ l, and the absorbance at 450 nm was measured via a plate reader (Dragon Wellscan MK3, LabSystems, Bucharest, Romania). The samples were measured on days 1, 3, 7, 14 and 21, and the results were normalized to the optical density (OD) measured on day 1.

2.3. *In vitro* osteogenesis analysis

2.3.1. Alkaline phosphatase (ALP) activity

Alkaline phosphatase (ALP) activity was determined by a colorimetric assay of enzyme activity (Abcam p-NPP Kit, Abcam). Briefly, after incubation for 7, 14 and 21 days in osteoinductive-factor-free NM and OIM, the bioprinted scaffolds were treated with 2 ml of 55 mM Na-citrate solution to decrosslink the bioink for releasing the embedded cells. The mixtures were centrifuged at 3000 rpm for 15 min. The supernatants were removed, 1 ml of lysis buffer was added, and the samples were incubated for 1 h. The lysates were sonicated and centrifuged again at 3000 rpm for 15 min, and the total protein content in the supernatant was quantified by the Bradford test (AppliChem GmbH, Germany). ALP activity of the encapsulated cells was quantified by a *p*-nitrophenyl phosphate (pNPP) colorimetric assay at 405 nm and normalized to the protein content.

2.3.2. *q*-rtPCR

The gene expression of Runx2, Osterix, ALP, Col 1a1, OCN and OPN was detected by qRT-PCR. The bioprinted scaffolds were cultured in NM or OIM *in vitro* for 7, 14 and 21 days before RNA was collected. A total RNA isolation kit (Takara, Xi'an, China) was used to extract RNA from the encapsulated cells. Table S2 shows the primers used for PCR. Total RNA was extracted as the template and reverse transcribed using a reverse transcription kit (Takara, Xi'an, China) for cDNA preparation. A quantitative real-time PCR detection system (Takara, Xi'an, China) was used, and *q*-rtPCR was performed with Fast Start Universal SYBR Green Master Mix (Takara, Xi'an, China) under the conditions of 95 °C for 10 min, 95 °C for 15 s and 60 °C for 1 min. PCR specificity was verified with the melting curve, and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) was used as an endogenous control in the $2^{-\Delta\Delta C_t}$ method to calculate the gene expression levels. The expression data were normalized to the expression levels on day 1. Each experiment was performed in triplicate.

2.3.3. Alizarin red staining

The mineral deposits of the laden-cells were visualized by Alizarin Red S staining. 3D bioprinted scaffolds ($n = 3$) were harvested after incubation for 14 and 21 days in NM and OIM. The preparation of paraffin sections was the same as described for H&E staining. After deparaffinization and rehydration, Alizarin Red S staining was performed according to the manufacturer's instructions, and images were captured using a microscope imaging system (BX53; Olympus, Japan). The 3D printed pure nanocomposite scaffolds without cells were used as controls. Then, the integrated optical density (IOD) was measured using Image Pro Plus (IPP) image analysis software (Version 6.0, Media

Cybernetics). The average optical density value = IOD/Area was calculated.

2.3.4. Microscopic analyses

To observe the formation of calcium nodules on the surface of the bioprinted scaffolds, SEM (S-3400; Hitachi, Tokyo, Japan) using a Bruker XFlash 5030 EDX detector was performed. After incubation *in vitro* for 14 days in NM, the bioprinted scaffolds were washed in deionized water three times to remove the soluble Ca^{2+} and phosphate, dehydrated through an ascending ethanol series (diluted with deionized water) ending in 100% ethanol and finally dried at room temperature in a vacuum. Elemental mapping and compositional analysis of calcium, phosphorus, silicon and magnesium were performed for all samples ($n = 3$) under identical conditions at an accelerator voltage of 20 keV and a pressure of 12 Pa.

2.4. *In vivo* implantations

2.4.1. Surgical procedure

One hundred and twenty Sprague Dawley rats (SD rats, male, 200–250 g) were purchased from the Laboratory Animal Center, Fourth Military Medical University (NO.61002000000913). All animal procedures were approved by and carried out in accordance with the standards of the Experimental Animal Ethics Committee of Fourth Military Medical University.

A rat critical-size calvarial defect model [49] was employed for the bone defect repair experiment, and all surgical procedures were completed under sterile conditions. One defect was created per animal, and six scaffolds were implanted per time point each group. The 120 rats were divided into five equal groups: a no scaffold (BLK) group, a 3D printed 0%*n*Si pure Gel-Alg scaffold (0%*n*SiPPS) group, a 3D printed 2%*n*Si pure nanocomposite scaffold (2%*n*SiPPS) group, a 3D bioprinted rBMSCs-laden 0%*n*Si Gel-Alg scaffold (0%*n*SiBPS) group, and a 3D bioprinted rBMSCs-laden 2%*n*Si nanocomposite scaffold (2%*n*SiBPS) group. Briefly, after anesthetization by intraperitoneal injection of pentobarbital sodium (100 mg kg^{-1}), each rat's head was shaved and disinfected with iodophor. A sagittal incision of approximately 15 mm was made on the scalp, and the calvarium was exposed by blunt dissection. A defect of 8 mm in diameter was made. One scaffold with an 8-mm diameter and a 1.5-mm thickness was inserted into the defect, and the defect was closed with 3-0 silk sutures. The scaffolds and the surrounding skull, hearts, livers, spleens, lungs, and kidneys were harvested at weeks 2, 4, 8 and 12 postimplantation. All samples were fixed in 4% paraformaldehyde for 2 days, and micro-CT and histological analyses were performed later.

To visualize new bone formation and mineralization, a polychrome sequential fluorescent labeling method was used. Two different fluorochromes, 20 mg kg^{-1} calcein and 30 ml kg^{-1} alizarin red S, were sequentially administered intraperitoneally at 14 and 4 days before euthanizing rats at the 8-week ($n = 3$) and 12-week ($n = 3$) time points.

2.4.2. Microcomputed tomography (micro-CT) analysis and histological evaluation

Micro-CT (ZKKS-MCT-SharpII, Zhongke Co., China) analysis was performed at a voltage of 60 kV and an electric current of 67 mA. The voxel size after reconstruction was $25 \times 25 \times 25 \mu\text{m}$. Based on the micro-CT results, 3D images were reconstructed by Materialise's Interactive Medical Image Control System (MIMICS, Materialise Co., Belgium). The bone volume to total volume (BV/TV) ratio at weeks 2, 4, 8 and 12 after surgery was analyzed.

Next, 50 μm non-decalcified sections were prepared using an Exakt system (model 310 CP band system, Exakt, Oklahoma City, OK, USA) to observed fluorescence labeling under a fluorescence microscope and to calculate the mineral apposition rate (MAR) of new bone formation, which refers to the distance between the two different labels ($\mu\text{m}/\text{d}$)

($n = 3$). In addition, Van Gieson staining was performed to visualize the mineralized bone tissue (pink) in the same sections. The images were photographed by a fluorescence microscope (BX53; Olympus, Japan).

The remaining non-decalcified samples were decalcified in 10% EDTA for 4 weeks, as well as the fixed hearts, livers, spleens, lungs, and kidneys, embedded in paraffin, and sectioned. Serial cross-sections of decalcified samples underwent H&E and Masson's trichrome staining according to the manufacturer's instructions. Then, images were acquired using a light microscope (BX53; Olympus, Japan) and a video camera (Soft Imaging System). Quantitative analysis was carried out using ImageJ software to show the relative amount of new bone formation. Three samples of every time point were selected and newly formed mineralized tissue area in each sample was calculated and shown as a percentage to total tissue area.

2.5. Statistical analysis

The data are presented as the mean $\pm$ standard deviation. One-way analysis of variance (ANOVA) or Student's *t*-test was used to analyze the significant differences between samples. For all tests, the level of significance was set to $*p < 0.05$, $**p < 0.01$ and $***p < 0.001$.

3. Results and discussion

Due to the poor mechanical strength of gelatin, GelMA is the most commonly used form of gelatin in 3D printing. However, the covalently cross-linked chemistries generally cross-react with functional groups present on the cell surface, which can potentially lead to some undesired effects [27]. The photo-cross-linked hydrogels usually involve the use of toxic initiators to induce the generation of free radicals for a cross-linking reaction by UV- or visible-light exposure, while such UV light, toxic initiators, or free radicals may damage the cells [28–31].

Additionally, to maintain the self-proliferation and differentiation characteristics of the laden- cells within the bioinks as far as possible, it is critical to mimic the natural extracellular matrix (ECM) comprising of fibrous structural proteins (e.g. collagen) and polysaccharides, with a complex 3D network. Within this network, collagen provides laden cells with mechanical resistance to tensile and shear stress, while polysaccharides are the water-absorbing components and provides laden cells with resistance to compressive forces [32,33].

Therefore, to mimic the natural ECM and explore a more convenient, low-cost, and biocompatible bioink to decrease the side effects of covalent crosslinking interactions of bioprinted products, we firstly developed a functional and biomimetic nanocomposite hydrogel composed of collagen derived gelatin, polysaccharide alginate and nSi (Fig. 1A, B).

3.1. Physical characterization of the nanocomposite hydrogel

We fabricated physical cross-linked nanocomposite hydrogel samples by reducing the temperature from 37 °C to 25 °C and immersing the samples in 2% w/v CaCl_2 solution.

Fourier transform infrared spectroscopy (FT-IR) (Fig. 1C) was then used to confirm the presence of nSi within the Gel/Alg network after physical cross-linking. The peak observed near 1000 cm^{-1} was due to stretching of Si-OH and Si-O and bending of Si-O-Si bonds in the nSi within the nanocomposite hydrogels. Alginate displayed a carboxyl peak at 1443 cm^{-1} and gelatin showed prominent amine peaks near 1540 and 1630 cm^{-1} . Overall, the nanocomposite hydrogel exhibited Si-OH, Si-O, Si-O-Si and carboxyl and amine peaks, indicating the presence of nSi, alginate and gelatin. Energy-dispersive X-ray spectroscopy (EDX) was performed to demonstrate the presence of silicon and magnesium in the nanocomposite hydrogels, and as these elements were not observed in the Gel-Alg hydrogel without nSi (Fig. 1D). Scanning electron microscopy (SEM) images also confirmed no obvious particle aggregation in the nanocomposite hydrogel (Fig. 1D).

Hydration properties of the hydrogels play a critical role in determining the applicability in biotechnological and biomedical applications. The diffusion of nutrients and waste products strongly depends on the hydration properties of printed materials, which is essential for the viability of loaded cells [34]. The hydration properties of Gel-Alg based hydrogels with different concentrations of nSi were evaluated in phosphate-buffered saline (PBS). The hydration degree decreased with an increase in nSi concentration when compared to pure Gel-Alg hydrogel (0% nSi) (Fig. S1). The swelling ratio is an indirect representation in terms of total water content. For example, 2% nSi networks with a swelling ratio of 9.6 contain $> 90\%$ of PBS, whereas the 0% nSi networks with a swelling ratio of 12.4 contain $> 92.5\%$ PBS, indicated that the nanocomposite hydrogels was sufficient to maintain high cell viability. However, highly hydrophilic hydrogels for 3D bioprinting exhibit extremely high swelling ratios, which can results in deformation of the printed constructs and significant loss of resolution [35]. With the increase of nSi concentration, the swelling ratio decreased, which also indicated that the addition of nSi may increase the resolution and fidelity of the 3D bioprinted scaffolds.

The porosity of bioinks also plays a vital role in directing cellular behavior, such as expansion and migration [28,36]. The size and distribution of pores can affect cellular infiltration and distribution throughout the hydrogel network [36]. The addition of nSi to the Gel-Alg mixture led to an increase in the pore size of the physical cross-linked network (Fig. S2). The increase in pore size may be ascribed to the electrostatic interactions between nSi and gelatin or alginate, which locally condensed the polymer network and resulted in the formation of larger voids [26].

3.2. Rheological and mechanical analysis of Laponite enhanced nanocomposite hydrogel

3.2.1. Enhancing the printability by Laponite

To investigate the printability and rheological characteristics of the six kind hydrogels (containing 10% gelatin, referred to as Gel; 1% alginate, referred as Alg; and 10% Gel-1% Alg-0%/1%/2%/3% nSi, referred to as 0%nSi, 1%nSi, 2%nSi, 3%nSi, respectively), the shear-thinning behavior and the yield point were examined at physiological temperature (37 °C). Fig. 2A shows that the prepolymer solutions (Gel, Alg and 0%nSi) were liquid at 37 °C, and the addition of nSi significantly increased the viscosity of the solutions. The 1%nSi, 2%nSi and 3%nSi solutions displayed clear shear-thinning behavior, as a significant decrease in viscosity was observed with increased shear rate. When the nSi concentrations was higher than 2%, the solutions resulted in a semisolid that is suitable for 3D printing. The shear stress-viscosity curve (Fig. 2B) showed that the hydrogels transited from a semisolid-like to liquid-like state (*i.e.*, yield stress). This phenomenon can be described by the Herschel-Bulkley fluid model, $\tau = \tau_0 + K(\dot{\gamma})^n$, where $\dot{\gamma}$ is the shear rate, and τ and τ_0 represent the apparent stress and the yield stress, respectively. The consistency index *K* is the apparent viscosity of the fluid at a shear rate of 1 s^{-1} [37]. Table S1 shows that the three formulations exhibited shear-thinning behavior at 37 °C ($n < 1$). The addition of nSi significantly decreased the *n* value (from 0.57 to 0.36), which confirmed that nSi increased the shear thinning character of the nanocomposite hydrogels. Thus, nSi can improve printability by reducing the flow resistance at a high shear rate generated during extrusion, which may protect the embedded cells from damage due to shear stress. Similarly, the addition of nSi significantly increased the values of both *K* (from 1.88 to 13.11 Pa) and τ_0 (from 18.7 to 133 Pa). Higher yield stresses allow bioprinted scaffolds to behave as a viscoelastic solid after extrusion, even before CaCl_2 solution cross-linking, which helps prevent collapse and increases fidelity.

Moreover, the effects of the addition of nSi to the Gel-Alg network and the 3D printability were investigated by oscillation stress sweep to calculate the elastic storage (G') and loss (G'') moduli at 37 °C (Fig. 2C). The non-crosslinked solution (0%nSi) was a liquid as there was no

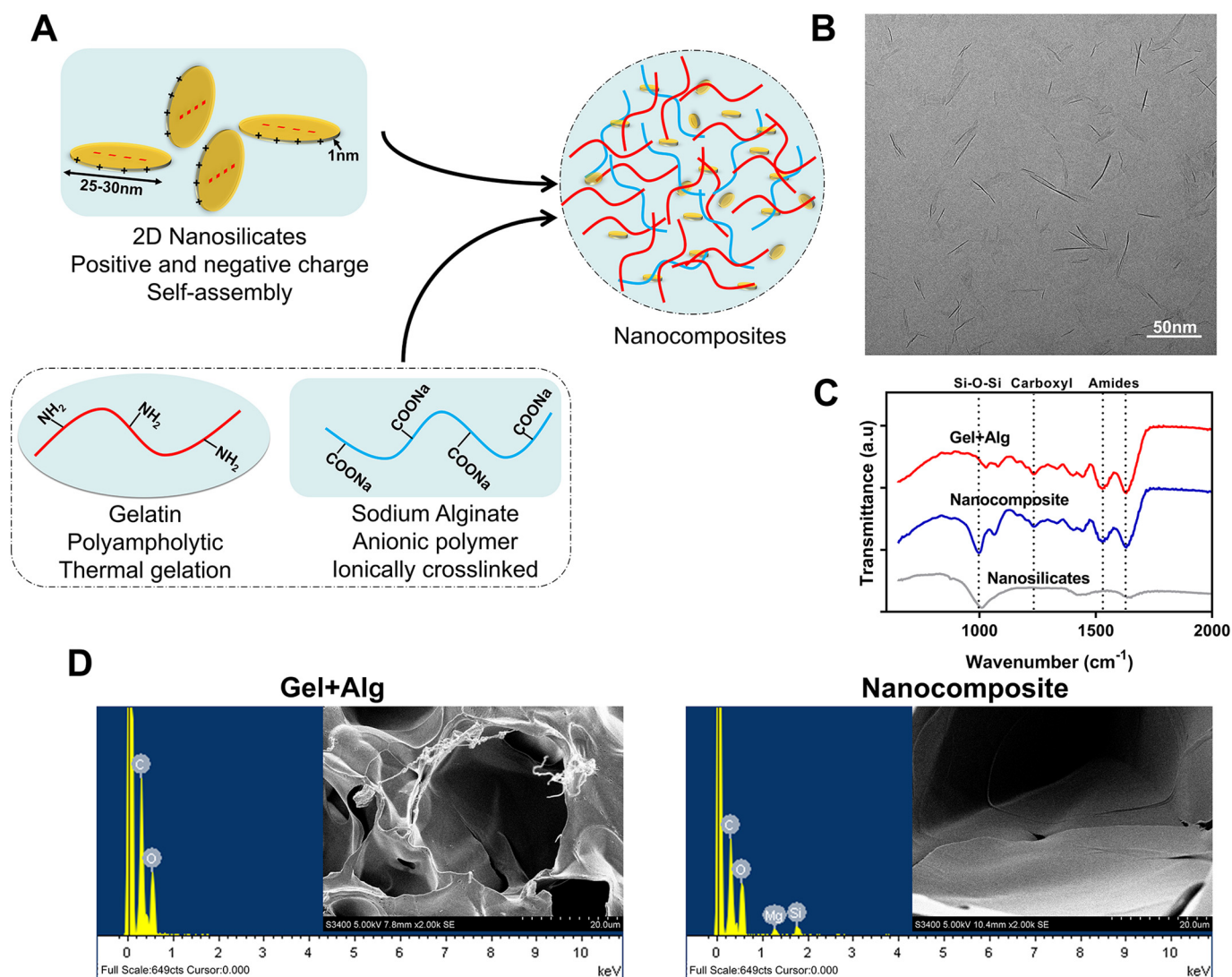


Fig. 1. Preparation and physical characterization of the nanocomposite hydrogels. A. Schematic of the preparation of the nanocomposite hydrogels. B. Cryo-TEM image of 2% w/v nSi suspension. C. FT-IR spectra of different hydrogels showed the presence of a Si-O-Si peak (1000 cm^{-1}), carboxyl peak (1443 cm^{-1}) and amides peak (1540 and 1630 cm^{-1}) in the nanocomposite hydrogel composed of Gel, Alg and nSi. D. The results of EDX measurements of Gel-Alg and the nanocomposite composed of Gel, Alg and nSi; the SEM images suggested that the nSi nanoparticles were uniformly distributed within the polymeric network with a lack of nanoparticle aggregation and phase separation. (nSi: nanosilicates; Gel: gelatin; Alg: alginate).

cross-point of the G' and G'' curve, which was in accordance with the shear stress-viscosity curve. Furthermore, the addition of nSi significantly increased the G' and G'' of the solution. The three cross-points represented the yield stress points which was in accordance with the shear stress-viscosity curve. When the temperature decreased to $25\text{ }^{\circ}\text{C}$ (Fig. 2D), the G' of 0% nSi solution was higher than G'' which means that the thermo-sensitivity of gelatin made it in a solid-like state, though the G' is very low. Even though the addition of nSi significantly increased the G' , the G' of 2%nSi and 3%nSi were nearly at the same level.

To sum up, the 2%nSi nanocomposite hydrogel not only showed an appropriate yield stress with an excellent shear thinning character, but also exhibited nearly the same level of G' with 3% nSi, indicated that it was more mild to protect the laden cells for 3D bioprinting process.

3.2.2. Enhancing the mechanical strength by Laponite

The mechanical properties of 3D bioprinted structures are most important for maintaining structural integrity and regulating cell fate and activity [7,38]. To evaluate the mechanical properties of the present nanocomposite hydrogels, uniaxial compression testing of four

types of hydrogel samples (0%, 1%, 2%, 3%nSi) was performed. The strain-stress curve indicated an increasing trend in stress with increasing strain (Fig. 2E). The addition of nSi corresponded to a significant increasing trend in the compressive modulus. The results showed that the nanocomposite hydrogels with 2% nSi ($101.41 \pm 3.11\text{ kPa}$) displayed a nearly 2.2-fold higher compressive modulus than Gel-Alg alone (0%nSi) ($46.98 \pm 1.54\text{ kPa}$) and nearly 2-fold higher compressive modulus than 1%nSi ($57.82 \pm 1.69\text{ kPa}$), and the nanocomposite hydrogels with 3% nSi ($123.59 \pm 2.10\text{ kPa}$) showed a nearly 2.6-fold higher compressive modulus than 0%nSi (Fig. 2F) ($***p < 0.001$). Furthermore, in contrast to other types of nanoparticles, such as gold nanorods incorporated within gelatin hydrogels, our bioink also displayed a greater increase in the compressive modulus [39], which further proved that nSi in our bioink not only acted as a filler but also increased the mechanical strength of the hydrogel network.

According to the rheological and mechanical analysis, we found that the addition of nSi could significantly increase the printability compared to Gel-Alg alone, and with the increased concentration of the nSi, the mechanical strength increased. Moreover, the 2%nSi showed nearly

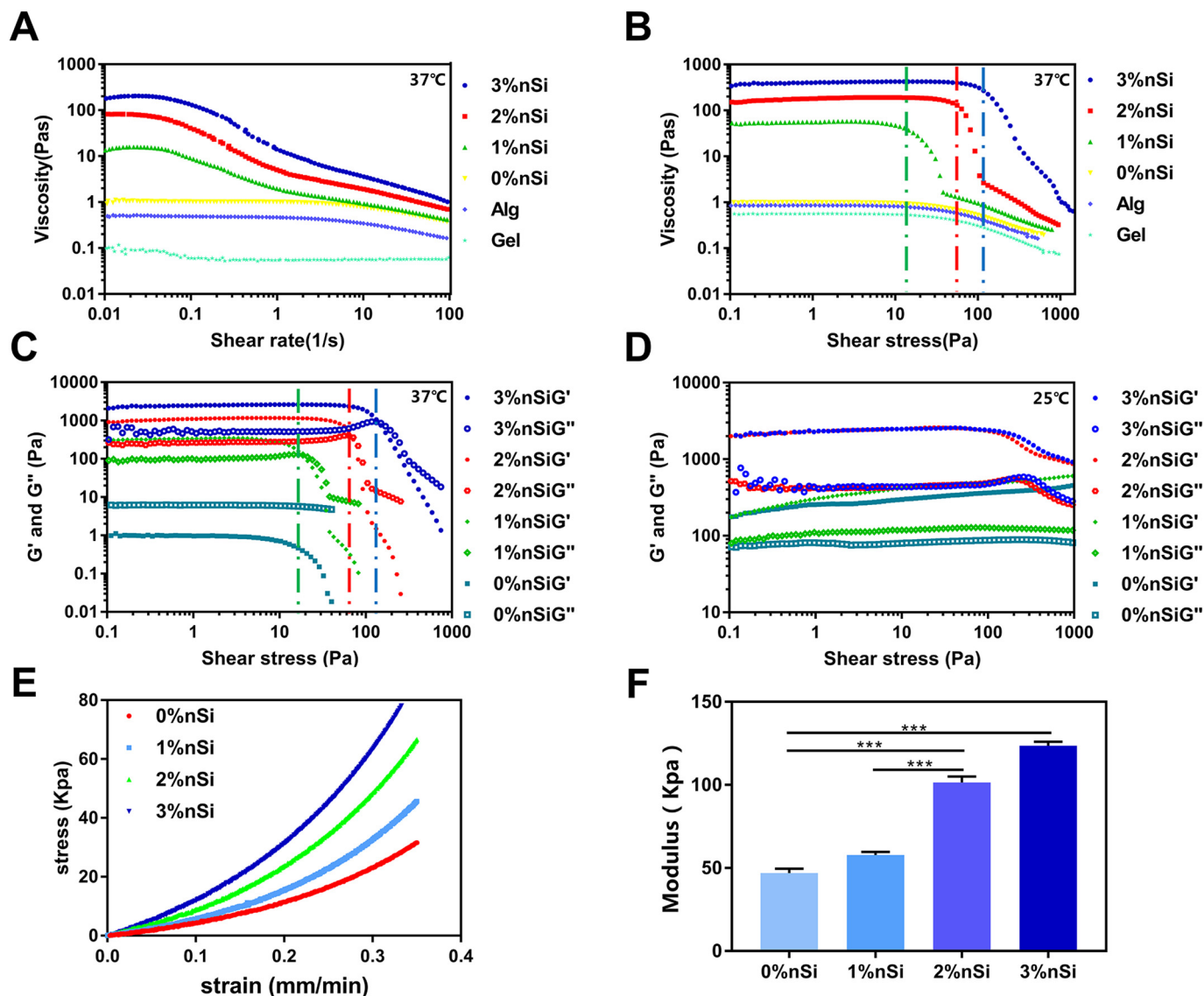


Fig. 2. Rheological and mechanical evaluation of the nanocomposite hydrogels. A. Shear rate sweep at 37 °C illustrated the shear thinning characteristics of pre-cross-linked gels. B. Shear stress sweeps at 37 °C show that the viscosity changed with increasing shear stress, and the dotted lines suggests the yield stress point of each hydrogel. C. Oscillation stress sweeps indicate shear stress-dependent changes in the elastic storage (G') and loss (G'') moduli at 37 °C. D. Oscillation stress sweeps indicate shear stress-dependent changes in the elastic storage (G') and loss (G'') moduli at 25 °C. E. Four types of hydrogels were subjected to uniaxial compression up to 0.4 strain. F. The compressive modulus of the hydrogels was calculated from the stress-strain curves at 0.05–0.10 strain ($***p < 0.001$). (nSi: nanosilicates; Gel: gelatin; Alg: alginate).

the same mechanical strength and G' at 25 °C, while its yield stress was nearly one third of the 3%nSi at 37 °C, which is milder for the laden cells to through the printing needle. So we finally chose the 2%nSi as our 3D bioprinted hydrogel.

Gelatin is maintained in a fluidic state above 30 °C, whereas it gels below 30 °C [11,31]. To maintain the high viability of loaded cells, we printed the nanocomposite bioink at 37 °C, which is above the gelation temperature of gelatin alone, onto a constant temperature-controlled panel kept at room temperature (25 °C). With this system, Laponite nanocomposite hydrogels were successfully used to print a centimeter-scale construct (15 mm high and 60 layers) with enhanced mechanical properties and printing fidelity (Fig. S3).

To further evaluate the printability and high fidelity of the hydrogels, 3D printed scaffolds without cells were observed by SEM. Images taken of the front and sides of the scaffolds indicated that the 2%nSi hydrogel exhibited good printability and that the connections between layers was excellent (Fig. S4).

3.3. Laponite nanocomposite bioink scaffold supports cell viability and proliferation *in vitro*

The Laponite nanocomposite scaffolds (2%nSi) (15 mm × 15 mm × 1.5 mm, 6 layers) loaded with rat bone marrow mesenchymal stem cells (rBMSCs) were biofabricated under aseptic conditions (Fig. 3A, Video S1). To evaluate the survival of the laden cells, live/dead cell staining was performed. Observations of the top and sides of the 3D reconstructions revealed many living cells (color green) in a uniform distribution (Fig. 3B). The top views also showed spreading of cells within the hydrogel. As Laponite and the red dye, ethidium homodimer-1, linked together due to electrostatic interactions, the Laponite wave structures within the hydrogel were stained red, consistent with the published literature [40]. The results demonstrated the good cytocompatibility of the bioink, and the 3D bioprinting process was mild enough to retain a high number of living cells after printing. The high cell viability was also related to the high hydration degree and the porosity of the nanocomposite hydrogels, which

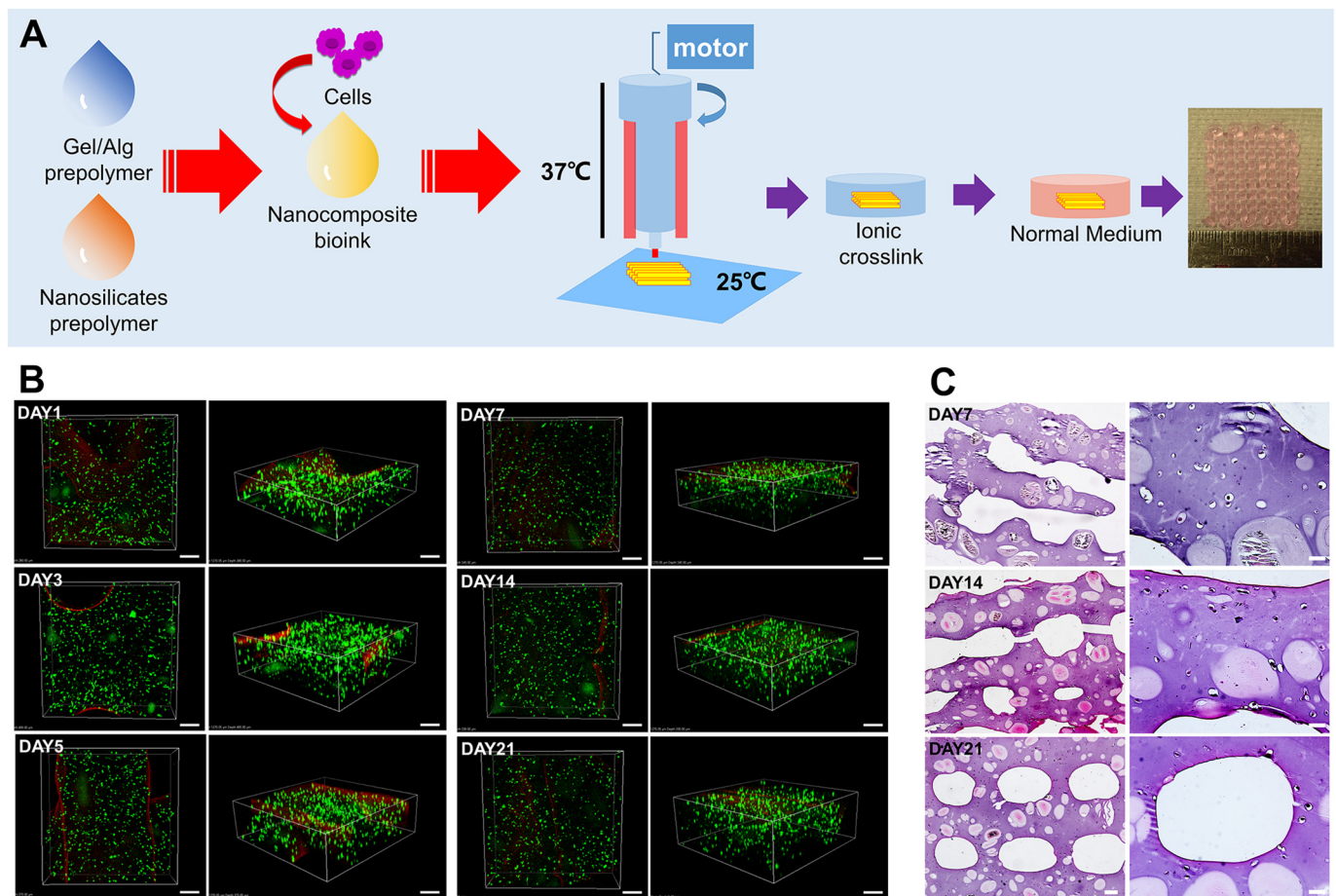


Fig. 3. The laponite nanocomposite bioink supports cell viability. **A.** Schematic of the 3D bioprinting process using a 2%nSi bioink. **B.** Live/dead cell staining of the 3D bioprinted scaffolds. 3D reconstruction of the confocal microscope images showed numerous live cells (stained green) uniformly distributed within the 3D bioprinted 2%nSi scaffolds after 1, 3, 5, 7, 14 and 21 days of culture in normal medium *in vitro*. Laponite nanoparticles were also stained red due to electrostatic interactions with ethidium homodimer-1 [40]. Scale bar: 100 μ m. **C.** H&E staining of sections of the bioprinted scaffolds (2%nSi) after 7, 14 and 21 days of culture in normal medium. Scale bar: left column: 200 μ m, right column: 50 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

provided an appropriate oxygenated and nutrient-rich microenvironment for the encapsulated cells.

Hematoxylin-eosin (H&E) staining was performed on sections of the 3D bioprinted structures (2%nSi) on days 7, 14 and 21 (Fig. 3C). The results further confirmed that the encapsulated cells were maintained at an appropriate density.

Co-staining of the cytoskeleton (red) and nuclei (blue) by rhodamine/phalloidin and DAPI was performed on day 7 *in vitro*. Fig. S5A shows the division (i), contacting (ii) and spreading (iii) of encapsulated cells, indicated that the cells remained alive well. Confocal 3D reconstructions further confirmed the uniform distribution of cells (Fig. S5B).

Proliferation of the encapsulated cells was determined by CCK-8 assays and the scaffolds without Laponite (0%nSi) (15 mm $\times$ 15 mm $\times$ 1.5 mm, 6 layers) loaded with rBMSCs were used as control (Fig. S5C). The results indicated that the laden cells maintained good viability and proliferated well at days 3, 7 and 14 *in vitro*, and the rBMSC proliferation in each group was similar at the three time point indicated that the additional nSi induced no apparent toxicity to the laden rBMSCs (Fig. S5D). To some extent, the high proliferation may be attributed to degradation of the polymers, which results in increased porosity and altered hydrogel elasticity [41].

3.4. Nanocomposite bioink scaffold enhances osteogenic differentiation and mineralization *in vitro*

In 2013, Khademhosseini et al. first reported Laponite-induced osteogenic differentiation of hMSCs in the absence of osteogenic supplements in 2D culture [25]. Thereafter, they developed an nSi and GelMA hydrogel platform for the osteogenic differentiation of hMSCs in 3D culture [42]. Considering the potential cytotoxicity and other undesired effects of covalent crosslinking induced by ultraviolet light, we designed a gelatin-based functional and biomimetic nanocomposite bioink to explore its ability to induce osteogenesis in normal medium after 3D bioprinting.

The differentiated osteoblasts from BMSCs were demonstrated by quantitative assays of alkaline phosphatase (ALP) activity, as upregulated ALP activity is also observed during the early phase of osteogenesis [25]. Fig. 4A shows that ALP activity significantly increased from day 7 to day 14 both in NM and OIM. Although ALP activity in the positive control group cultured in OIM (36.26 ± 2.63) was almost 1.2-fold higher than the in NM group (30.38 ± 1.75) at day 7 ($*p < 0.05$), the two groups exhibited almost the same level of ALP activity at day 14; ALP activity was 56.06 ± 2.14 in the NM group and 58.69 ± 3.01 in the OIM group. After cultured for 21 days, the ALP activity of the two groups was almost at the same level compared to day 14, and the two groups showed no difference; ALP activity was 60.06 ± 1.94 in the NM group and 62.70 ± 2.78 in the OIM group.

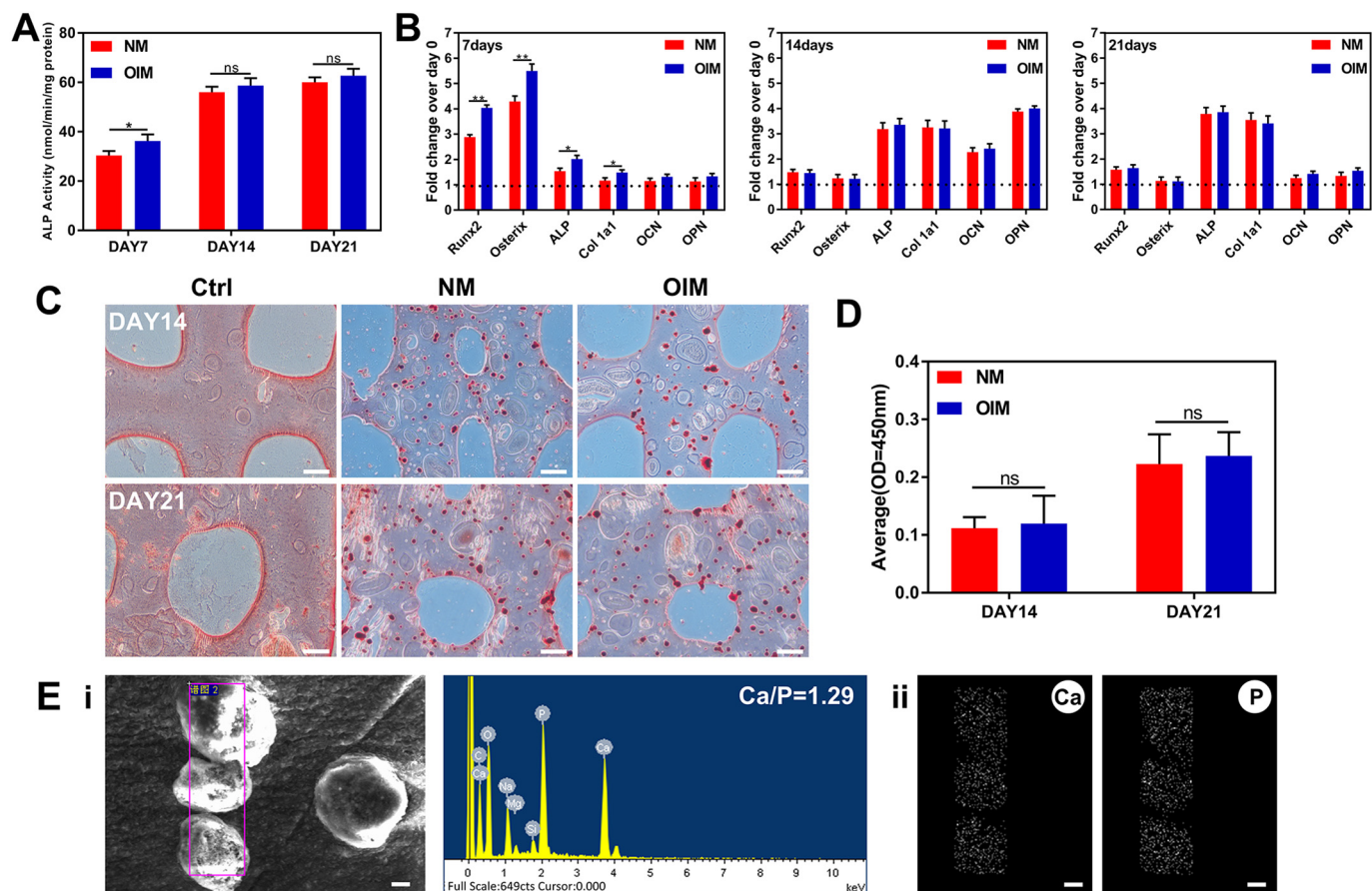


Fig. 4. The nanocomposite enhances osteogenic induction and mineralization *in vitro*. **A.** Quantitative assays of ALP activity of the encapsulated rBMSCs within the 3D bioprinted 2%Si scaffolds after culturing in normal medium (NM) or osteoinductive medium (OIM) for 7, 14 and 21 days. (* p < 0.05). **B.** q-rtPCR analysis of the encapsulated rBMSCs within 2%Si scaffold after culturing in NM or OIM for 7, 14 and 21 days. Gene expression was normalized to day 0. **C.** Alizarin Red S staining of sections of the 3D bioprinted 2%Si scaffolds after culturing in NM or OIM for 14 and 21 days *in vitro* showing the amount of the mineralized matrix deposited. 3D printed 2%Si scaffolds without cells were used as a negative control (Ctrl). Scale bar: 200 μ m. **D.** Quantitative analysis of the NM and OIM groups of positive staining for calcium nodules. **E. i)** Representative EDX mapping of calcium nodules on the surface of the 3D bioprinted rBMSC-laden 2%Si scaffolds cultured in NM for 14 days. Scale bar: 5 μ m. **ii)** The distribution of calcium and phosphorus elements, which is consistent with the nodules. Scale bar: 5 μ m. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The osteogenesis ability of embedded rBMSCs cultured in NM may be attributed to the osteoinductive character of the degradation products of nSi, which could be regarded as an osteoinductive agent [25]. With the increased degradation of nSi, the osteogenesis ability increased. As the degradation progress needs time, the ALP activity of NM group is lower than the OIM group at day 7. With the extending of incubation time, the ALP activity was nearly at the same level. Overall, the results indicated that nSi within the scaffolds upregulated ALP activity in the absence of osteoinductive factors.

Next, we tested the relative gene expression of RUNX2, Osterix, ALP, Col 1a1, OCN and OPN of the encapsulated cells cultured in NM and OIM for 7, 14 and 21 days (Table S2). RUNX2 belongs to the RUNX family of transcription factors and is the earliest and most specific marker of osteogenesis [43]. As shown in Fig. 4B, on the one hand, the RUNX2 and Osterix genes exhibited high expression at day 7 and decreased expression at day 14 and 21 both in NM group and OIM group; on the other hand, the RUNX2 expression is up-regulated nearly 3-folds after 7 days cultivation in NM compared with day 0, while nearly 4-folds was shown in OIM group (* p < 0.01); the Osterix expression is up-regulated > 4-folds after 7 days cultivation in NM compared with day 0, while nearly 5.5-folds was shown in OIM group (* p < 0.01). Both of the two genes showed the same level of expression at day 14 and 21. Furthermore, the expression of ALP and Col 1a1 genes increased from day 7 to day 14, and then maintained to day 21 for the

two groups. Especially, at day 7, the expression of the two genes in OIM is higher than that in NM (* p < 0.05), and then both of the two genes showed the same level of expression at day 14 and 21. In addition, the gene expression of ALP was also in accordance with the quantitative ALP activity assays. Moreover, OCN is the most abundant bone-specific noncollagenous protein synthesized by osteoblasts and serves as a marker for evaluating osteogenic maturation and bone formation [44]. OPN is a structural protein synthesized by preosteoblasts, osteoblasts, and osteocytes and is considered an important factor in bone remodeling [45]. The expression of the two genes were significantly increased from day 7 to day 14, and then decreased to day 21 in the two groups, and the expression of the two genes in the two groups exhibited no difference, further demonstrating that the nanocomposite hydrogels can significantly induce osteogenesis without any osteoinductive factors. Generally, as for the traditional tissue engineering scaffold where cells were planted on its surface, the qRT-PCR also showed that the Runx2 peaked after 14 days cultured *in vitro*, and declined on the 21st day, while its downstream effector proteins OCN and OPN peaked on the 21st day [46]. However, to some osteoinduction enhancement scaffolds where cells were planted on their surface, the OCN and OPN peaked on the 14th day *in vitro* [47]. Khademhosseini et al. have already proved that the addition of nSi could enhance the expression of OCN and OPN proteins *in vitro*. Our 3D bioprinted functional and biomimetic bioink scaffold could mimic the natural ECM and supply a more suitable

microenvironment for the encapsulated cells, and the exist of nSi could accelerate and enhance the osteoinduction of the rBMSCs. Therefore, the expression of OCN and OPN genes peaked at day 14 and showed a decrease at day 21. These results further reinforce the statement that the nSi can act as a promoter of osteogenic differentiation, by enhancing the protein production and its subsequent mineralization [25].

M. J. Coelho [48] demonstrated that NM is only a maintenance medium that does not trigger the osteogenic differentiation of hMSCs. And the addition of either ascorbic acid or dexamethasone still failed to induce formation of calcium phosphate deposits. Hence, to further evaluate the efficiency of ECM mineralization of the encapsulated cells in NM and OIM, sections of the 3D bioprinted structures were stained with Alizarin Red S as a marker for inorganic calcium. Fig. 4C shows the formation of a mineralized matrix in NM and OIM on days 14 and 21. The effect of nSi on nodule formation was obvious both in NM and OIM group. The laden rBMSCs exhibited the same ECM mineralization in the two groups, as shown by the quantitative analysis (Fig. 4D). To verify that Alizarin Red S-positive staining was due to mineral deposition, EDX and SEM analyses were performed to observe the mineralized calcium nodules. After culturing in NM for 14 days, EDX revealed strong signals corresponding to the bone-prevalent elements Ca, P, C, O, and Si and Mg on the surface of the nodules (Fig. 4E(i)). The ratio of Ca/P in the calcium nodules was (1.3 ± 0.21) , which is similar to the Ca/P ratio of hydroxyapatite, the main inorganic component of bone ($\text{Ca/P} \approx 1.67$), thus indicating effective mineralization of the nanocomposite bioink (Table S3). The distribution of calcium and phosphorus elements is consistent with that of nodules (Fig. 4E(ii)). Additionally, the SEM observations of the same area were in accordance with the EDX results (Fig. S6).

Overall, these data clearly suggest that our 3D bioprinted nanocomposite scaffolds can promote osteogenic differentiation of rBMSCs by enhancing specific gene or protein expression, subsequently increasing scaffold mineralization. The advantages of incorporating nSi include increased printability, mechanical strength and fidelity as well as the continuous induction of osteogenesis without the need for other expensive osteoinductive factors.

3.5. Laponite enhances bone healing *in vivo*

Most recently reported bioprinted constructs must be cultured in osteogenic medium *in vitro* for a long time (approximately 10–14 days) before they can be transplanted into bone defects [49], which may lead to additional difficulties in scaffold preparation and risks associated with contamination and limited activity of cells. MSCs were also reported to achieve limited bone formation *in vivo* due to a lack of proliferation upon prolonged *in vitro* differentiation [50]. In our study, we sought to evaluate the ability of our 3D bioprinted scaffolds to induce osteogenic differentiation and new bone formation *in vivo*. Thus, we transplanted the scaffolds cultured in normal media into critical-size cranial defects in Sprague Dawley (SD) rats within 24 h after bioprinting, without prior osteoinductive differentiation *in vitro*.

Critical-size full-layer defects of the cranium (Fig. S7) were established as previously described [51]. No infection and weight loss were observed during the experimental period, and the activity of the rats after the implantation was normal. The time points of 2 weeks, 4 weeks, 8 weeks and 12 weeks were chosen to evaluate bone healing in the five groups: a blank control group without scaffold (BLK), a 3D printed 0% nSi pure Gel-Alg scaffold (0%SiPPS) group, a 3D printed 2% nSi pure nanocomposite scaffold (2%SiPPS) group, a 3D bioprinted rBMSCs-laden 0% nSi Gel-Alg scaffold (0%SiBPS) group, and a 3D bioprinted rBMSCs-laden 2% nSi nanocomposite scaffold (2%SiBPS) group.

First, micro-computed tomography (micro-CT) was applied to measure the new bone mass of the defects. Fig. 5A shows the 3D reconstruction of typical samples; new bone is marked by the yellow circle (8 mm in diameter). The top view clearly shows that the 0% or 2%SiBPS group exhibited better bone formation than the 0% or 2%

SiPPS group, respectively; and the four groups exhibited significantly better bone formation than the BLK group. In particular, the 2%SiPPS group showed the same bone healing effect as the 0%SiBPS group, and the 2%SiBPS group showed the best bone healing effect in the five groups. A quantitative analysis (Fig. 5B) was performed to further evaluate the observed results. By weeks 2 and 4, the bone volume/tissue volume (BV/TV) ratio was $18.83 \pm 0.84\%$ and $20.30 \pm 0.57\%$ in the 2%SiBPS group respectively, but only $10.89 \pm 0.73\%$ and $12.84 \pm 0.65\%$ in the 2%SiPPS group, and $6.23 \pm 0.72\%$ and $8.21 \pm 0.85\%$ in the BLK group ($***p < 0.001$); the BV/TV ratio of the 0%SiBPS group was $10.08 \pm 0.73\%$ and $12.63 \pm 0.67\%$, and $7.29 \pm 0.53\%$ and $10.04 \pm 0.75\%$ in the 0%SiPPS group ($***p < 0.001$). At week 2, the 2%SiBPS group exhibited almost 2-folds more new bone formation than the 2%SiPPS group and the 0%SiBPS, and nearly 3-folds than the 0%SiPPS, indicating strong promotion of bone healing in the early stage in the presence of rBMSCs and nSi. By week 8, new bone filled $22.67 \pm 0.76\%$ of the defects in the 2%SiBPS group and $14.65 \pm 0.79\%$ of the defects in the 0%SiBPS, but only $8.63 \pm 0.88\%$, $14.25 \pm 0.51\%$ and $13.42 \pm 0.66\%$ of the defects in the BLK group, the 2%SiPPS group and the 0%SiPPS, respectively ($**p < 0.01$). By week 12, the bone healing process slowed down in the BLK, 0%SiPPS, 2%SiPPS and 0%SiBPS groups, as only another 2.44%, 3.11%, 3.85%, 4.12% of the defects were filled in these four groups, respectively, while an additional 7.15% of the defects was filled in the 2%SiBPS group, compared with week 8. Combined with the 3D reconstruction observations, the 2%SiBPS group exhibited the greatest promotion and acceleration of bone healing and repair than the other groups. Interestingly, in the bone healing progress, the 2%SiPPS group showed the same level of new bone formation as the 0%SiBPS group, indicated that the addition of nSi only may achieved the similar bone healing effect compared with the rBMSCs only.

Subsequently, sequential fluorescent labeling [51] was used to observe new bone formation by applying two types of fluorochromes: calcein (green) and alizarin red (red). Because fluorochromes can bind to calcium ions of newly formed bone and become incorporated into the site of the mineralization area, the new bone formed at different periods can be discriminated by different fluorochrome colors. Comparing the 2%SiBPS group with the 2%SiPPS, 0%SiBPS, 0%SiPPS and BLK groups, the largest interval between the two types of fluorescent signals was observed in the 2%SiBPS group, which indicated the fastest process of new bone formation both at week 8 and week 12 post-implantation (Fig. 6A). A quantitative analysis was then performed to further evaluate the mineral apposition rate (MAR). As shown in Fig. 6B, the MAR of the 2%SiBPS group were $(14.05 \pm 0.89) \mu\text{m/day}$ and $(14.47 \pm 0.58) \mu\text{m/day}$ at 8 weeks and 12 weeks postimplantation, respectively; while in the 2%SiPPS and 0%SiBPS groups, the MAR were $(7.65 \pm 0.86) \mu\text{m/day}$ and $(8.30 \pm 0.74) \mu\text{m/day}$, $(7.96 \pm 0.73) \mu\text{m/day}$ and $(8.57 \pm 0.82) \mu\text{m/day}$ at 8 weeks and 12 weeks postimplantation, respectively ($***p < 0.001$). The MAR of the last two groups shows no difference, indicated that the addition of nSi only may achieved the similar bone healing effect compared with the rBMSCs only, which was in accordance with the Micro-CT results. The MAR of both groups at 8 and 12 weeks were significantly higher than those of the BLK group of $(2.19 \pm 0.68) \mu\text{m/day}$ and $(2.99 \pm 0.57) \mu\text{m/day}$, respectively ($***p < 0.001$, $**p < 0.01$, $*p < 0.05$), and the 2%SiBPS group showed the fastest MAR, which indicated the superiority of bone healing of the 3D bioprintable nanocomposite bioink scaffolds. After that, Van Gieson staining was performed. Fig. S8 showed that the 2%SiBPS group exhibited the maximum amount formation of new bone, compared with the other four groups at 8 weeks. Moreover, after implanted for 12 weeks, a completed bony bridge was formed in the 2%SiBPS group. At the same time, the 0%SiBPS and 2%SiPPS groups also showed the formation of new bone, which is in keeping with the Micro-CT results.

H&E and Masson's trichrome staining were performed after decalcification to further evaluate new bone formation. H&E staining

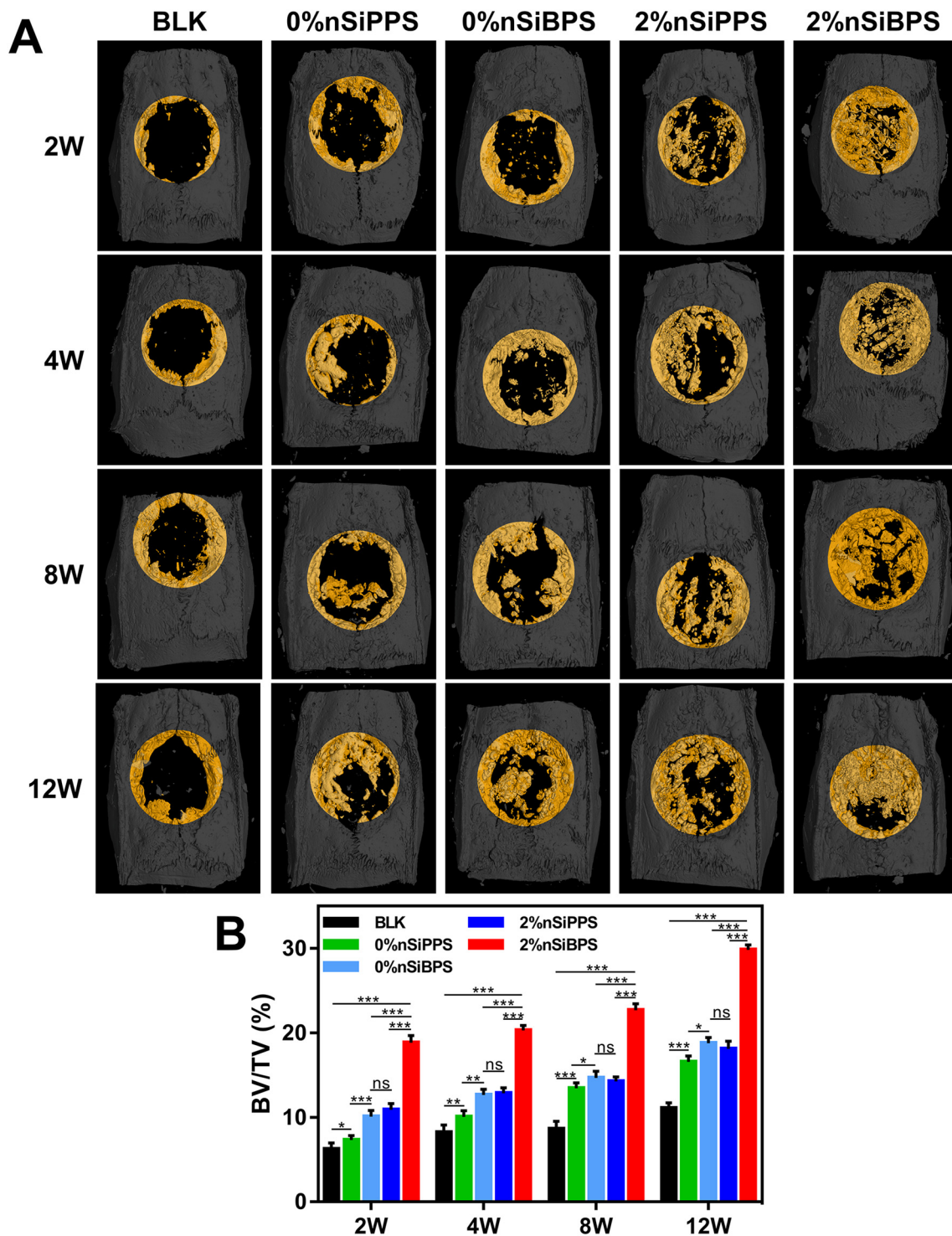


Fig. 5. Micro-CT analysis of bone healing. A. Micro-CT 3D reconstruction of new bone formation at 2, 4, 8 and 12 weeks after surgery. The yellow circle is 8 mm in diameter. B. Quantitative analysis of the bone volume/tissue volume (BV/TV) ratio of the five groups at different time points ($***p < 0.001$, $**p < 0.01$, $*p < 0.05$). (A blank control group without scaffold referred to as BLK, a 3D printed 0%nSi pure Gel-Alg scaffold group referred to as 0%nSiPPS, a 3D bioprinted rBMSCs-laden 0%nSi Gel-Alg scaffold group referred to as 0%nSiBPS, a 3D printed 2%nSi pure nanocomposite scaffold group referred to as 2%nSiPPS, and a 3D bioprinted rBMSCs-laden 2%nSi nanocomposite scaffold group referred to as 2%nSiBPS). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(Fig. 7) showed that the 2%nSiBPS group displayed extensive bone regeneration at 8 weeks. A bony bridge began to form, while some connective tissue remained on the side of the defect. Many new blood vessels (Fig. S9, marked by yellow arrows) existed around the newly formed trabeculae aligned by osteoblasts, suggesting active new bone formation, demonstrating a better capability of the functional and

biomimetic nanocomposite bioink to induce angiogenesis. As blood vessels play a critical role in the new bone formation process, the existence of more new blood vessels further confirmed the excellent bone healing ability in the 2%nSiBPS group. However, defects in the BLK group were mainly filled by dense granulation tissue. Furthermore, the area of the new bone formation in the 0%nSiPPS, 0%nSiBPS and 2%

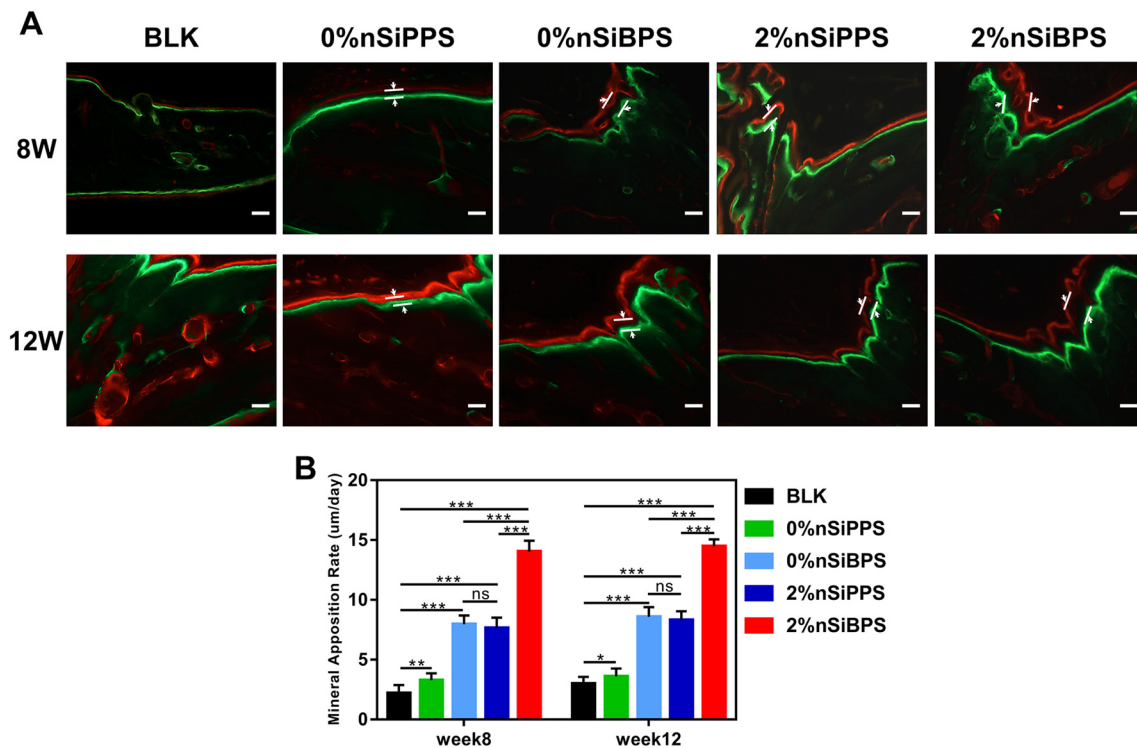


Fig. 6. A. Representative sequential fluorescent labeling at 8 and 12 weeks after surgery for the five groups. Scale bar: 100 μm. B. Quantitative analysis of the mineral apposition rate (μm/day) for sequential fluorescent labeling (**p < 0.001, *p < 0.01, *p < 0.05). (A blank control group without scaffold referred to as BLK, a 3D printed 0%nSi pure Gel-Alg scaffold group referred to as 0%nSiPPS, a 3D bioprinted rBMSCs-laden 0%nSi Gel-Alg scaffold group referred to as 0%nSiBPS, a 3D printed 2%nSi pure nanocomposite scaffold group referred to as 2%nSiPPS, and a 3D bioprinted rBMSCs-laden 2%nSi nanocomposite scaffold group referred to as 2%nSiBPS).

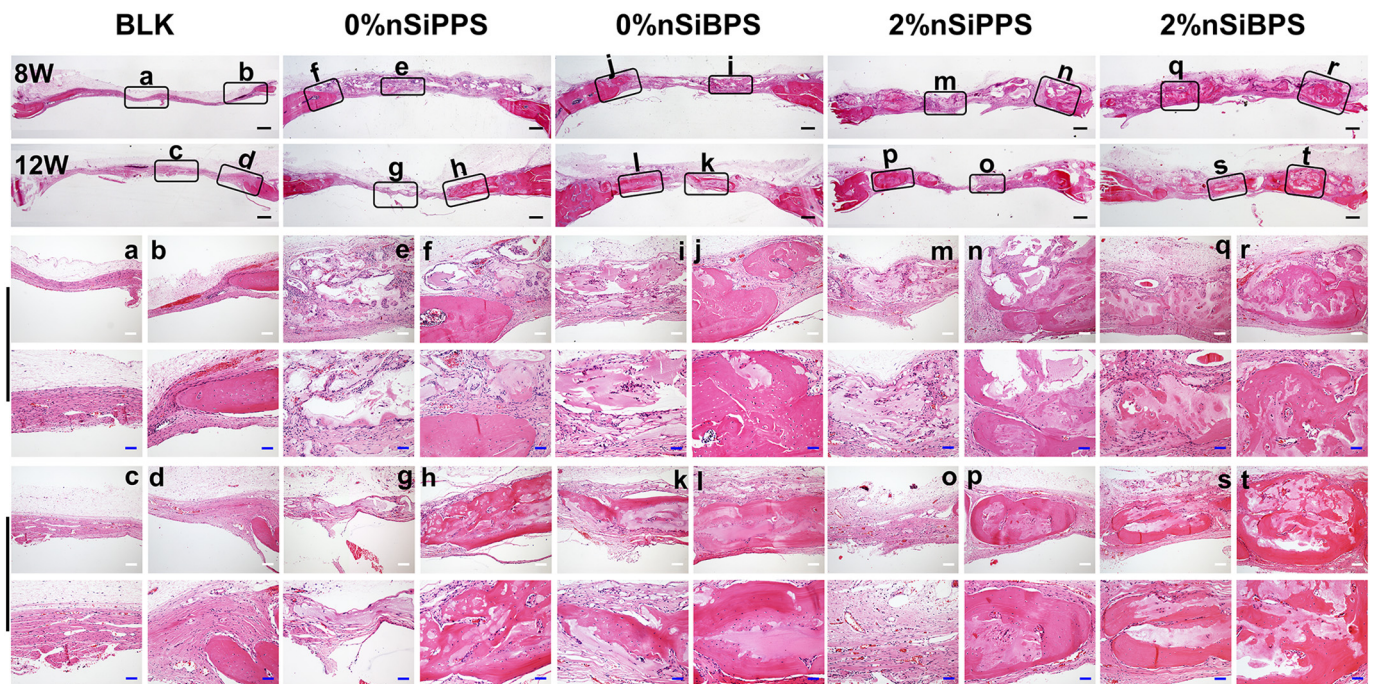


Fig. 7. H&E staining of the decalcified sections. a, c) Middle of the defect in the blank group (BLK). b, d) Margin of the defect in the BLK group. e, g) Middle of the defect in the 3D printed 0%nSi pure Gel-Alg scaffold (0%nSiPPS) group. f, h) Margin of the defect in the 0%nSiPPS group. i, k) Middle of the defect in the 3D bioprinted BMSCs-laden 0%nSi Gel-Alg scaffold (0%nSiBPS) group. j, l) Margin of the defect in the 0%nSiBPS group. m, o) Middle of the defect in the 3D printed 2%nSi pure nanocomposite scaffold (2%nSiPPS) group. n, p) Margin of the defect in the 2%nSiPPS group. q, s) Middle of the defect in the 3D bioprinted BMSCs-laden 2%nSi nanocomposite scaffold (2%nSiBPS) group. r, t) Margin of the defect in the 2%nSiBPS group. Scale bar: black: 500 μm, white: 100 μm, blue: 50 μm. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

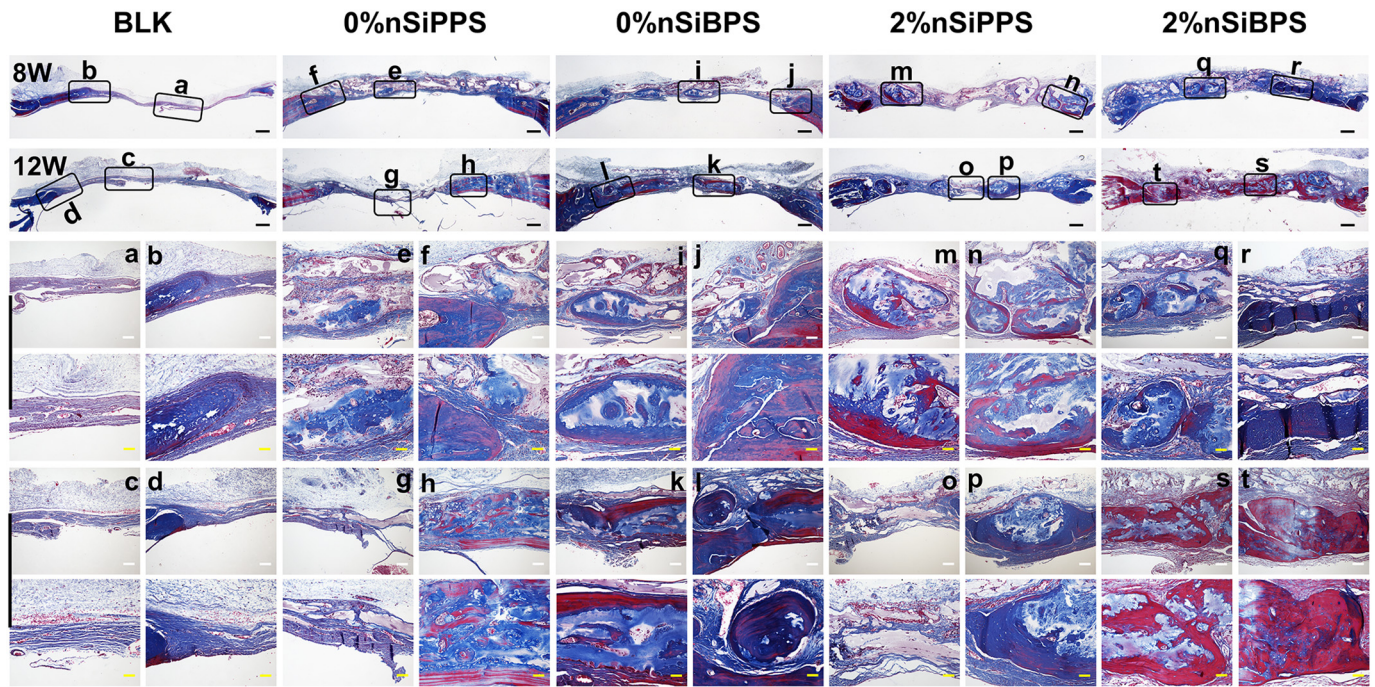


Fig. 8. Masson's trichrome staining of the decalcified sections. a, c) Middle of the defect in the blank group (BLK). b, d) Margin of the defect in the BLK group. e, g) Middle of the defect in the 3D printed 0%nSi pure Gel-Alg scaffold (0%nSiPPS) group. f, h) Margin of the defect in the 0%nSiPPS group. i, k) Middle of the defect in the 3D bioprinted BMSCs-laden 0%nSi Gel-Alg scaffold (0%nSiBPS) group. j, l) Margin of the defect in the 0%nSiBPS group. m, o) Middle of the defect in the 3D printed 2%nSi pure nanocomposite scaffold (2%nSiPPS) group. n, p) Margin of the defect in the 2%nSiPPS group. q, s) Middle of the defect in the 3D bioprinted BMSCs-laden 2%nSi nanocomposite scaffold (2%nSiBPS) group. r, t) Margin of the defect in the 2%nSiBPS group. Scale bar: black: 500 μm , white: 100 μm , yellow: 50 μm . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

nSiPPS groups were much lower than that in the 2%nSiBPS group. At 12 weeks postimplantation, in the 2%nSiBPS group, bony bridges appeared to connect defect margins, and many more mature bone structures could be identified. Cortical-like lamellar bone combined with woven bone, bone marrow space and many blood vessels appeared. In contrast, very minimal new bone formation was observed in the margin area of defects in the BLK. The defects were filled with 0%nSiPPS, 0%nSiBPS and 2%nSiPPS exhibited cortical-like lamellar bone formation, while the new bone mass was much lower than that in the 2%nSiBPS group (Fig. S10A).

Masson's trichrome staining further confirmed new bone formation and maturation (Fig. 8). In the BLK group, defects mainly displayed blue-stained fibrous tissues. Newly formed woven bone (blue-stained) filled larger areas in the defects in the 2%nSiBPS group than in the 0%nSiPPS, 0%nSiBPS and 2%nSiPPS groups at week 8. Additionally, at week 12, more arranged bone structures exhibiting cortical-like lamellar bone stained red and woven bone stained blue were clearly observed in the 2%nSiBPS group. This was further confirmed by a quantitative analysis of the bone formation areas, showing the relative amount of new bone in the calvarial defect area from different groups (Fig. S10B).

Together, the results sufficiently demonstrated the efficient bone healing ability of the 3D bioprinted functional and biomimetic nanocomposite bioink scaffold *in vivo* during both the early stage of bone formation and the late stage of bone maturation. This result may partly be due to the degradation products of the nanocomposite scaffolds, the bioactive degradation products, such as Mg^{2+} , orthosilicic acid, and Li^+ , have been demonstrated to promote bone formation *in vivo* [52]. As the 2%nSiPPS group also exhibited relatively excellent bone healing ability compared with other groups. Furthermore, the rBMSCs embedded within the 2%nSiBPS have the ability to undergo osteogenic differentiation, as demonstrated in previous *in vitro* studies. Additionally, H&E staining demonstrated that the encapsulated rBMSCs survived excellently *in vivo* after 2 weeks of implantation (Fig. S11),

which guaranteed the further osteogenic differentiation capability of the rBMSCs. To confirm the long-term biosafety of the four kinds of scaffolds (0%nSiPPS, 0%nSiBPS, 2%nSiPPS and 2%nSiBPS), H&E staining of rat heart, liver, spleen, lung, and kidney after implantation for 12 weeks was performed. No obvious histopathological abnormalities, *i.e.* no cell necrosis or damage, were found in the observed organizations (Fig. 9), clearly demonstrating that the scaffolds did not cause systemic toxicity to recipients [53–55].

Recent studies have also shown that the paracrine effects of MSCs in response to the local microenvironment play a critical role in injured tissue [56]. Among these paracrine effects, modulation of the switch of the macrophage to M2 phenotype and the beneficial reprogramming events play particularly crucial roles in tissue engineering [57]. Hence, in our study, in addition to the osteogenic capability of rBMSCs themselves, these paracrine effects may also play critical roles in the osteoimmunomodulation of the osteoinductive effects of Laponite, as the 0%nSiBPS group showed nearly the same bone healing level with the 2%nSiPPS group, which also further indicated the osteoinductive effect of nSi *in vivo*. Li et al. [58] showed that rBMSCs could reverse the phenotype switch of macrophages induced by Laponite, inhibit proinflammatory cytokines, and promote the production of anti-inflammatory factors and osteo-related cytokines *in vitro* and *in vivo*. Therefore, considering the osteoimmunomodulatory effects of rBMSCs, paracrine effects may be another mechanism for the progression of osteogenesis in 0% or 2%nSiBPS groups, which requires a deeper understanding in the future.

Laponite was initially applied by several researchers in drug and bioactivator delivery systems. Tiago B. Becher et al. [59] recently described a new drug delivery platform based on Laponite nanodiscs. Moreover, we avoided covalent modification of gelatin and only used CaCl_2 solution for crosslinking in our nanocomposite; thus, the natural affinity towards many proteins and the high activity of the nanocomposites were retained [60]. Hence, our Laponite nanocomposite hydrogel can be used not only as a bioink for 3D bioprinting in tissue

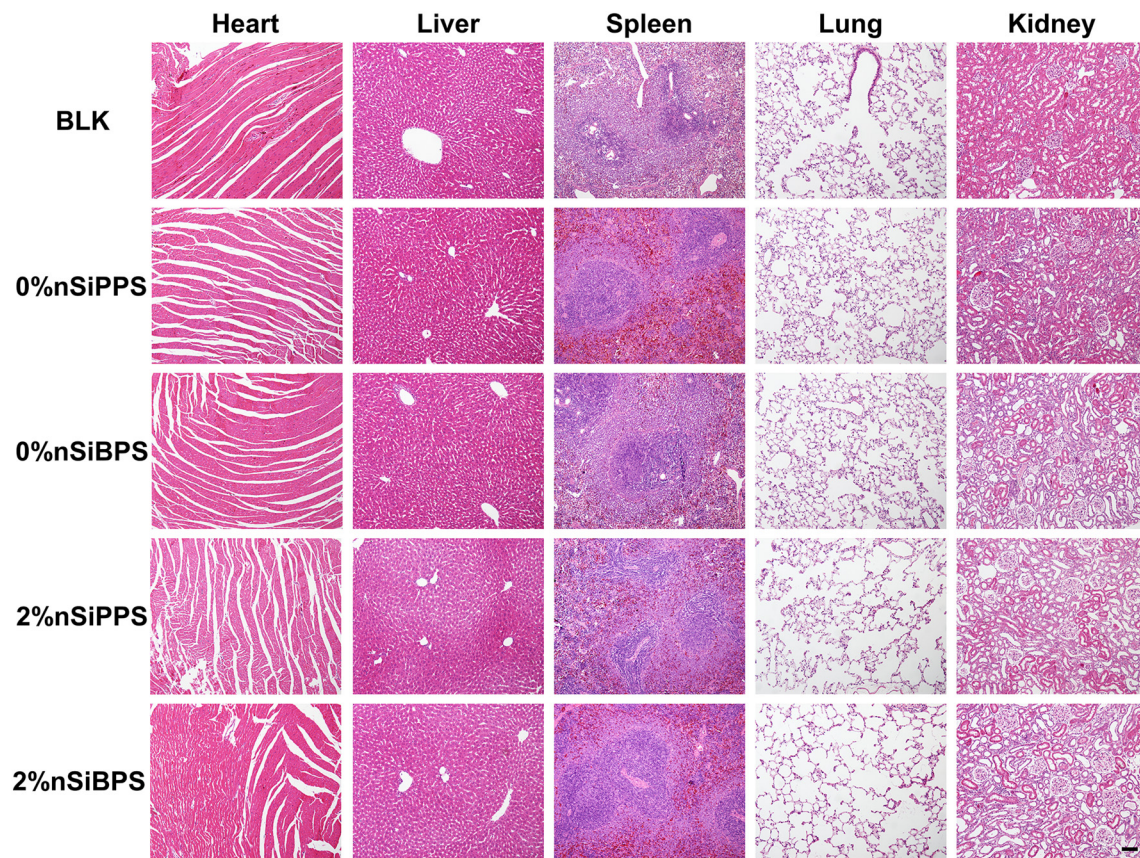


Fig. 9. H&E staining of rat heart, liver, spleen, lung, and kidney after implantation of four kinds scaffolds for 12 weeks. No obvious histopathological abnormalities in all observed organizations. Scale bar: 100 μ m. (A blank control group without scaffold referred to as BLK, a 3D printed 0%nSi pure Gel-Alg scaffold group referred to as 0%nSiPPS, a 3D bioprinted rBMSCs-laden 0%nSi Gel-Alg scaffold group referred to as 0%nSiBPS, a 3D printed 2%nSi pure nanocomposite scaffold group referred to as 2%nSiPPS, and a 3D bioprinted rBMSCs-laden 2%nSi nanocomposite scaffold group referred to as 2%nSiBPS).

engineering but can also be applied for drug controlled release or screening and the delivery of bioactive factors, such as VEGF, which expands its range of applications.

4. Conclusions

In conclusion, we developed and characterized a 3D bioprintable functional and biomimetic nanocomposite bioink for 3D bioprinting of functional tissue-engineered bone constructs. We demonstrated that the addition of nSi to the bioink not only improved its printability, especially along the Z-axis, but also increased its mechanical strength, making it possible to develop human-scale functional tissues or organs. Moreover, the bioink was confirmed to be both cytocompatible and osteoinductive for the laden rBMSCs without the incorporation of osteogenic factors in the bioprinted constructs. The safety and rapid bone healing capability of the 3D bioprinted nanocomposite scaffolds were further demonstrated *in vivo* in a rat critical-size (8 mm) bone defect model, which providing a clinical potential candidate for orthopedic applications. As Laponite also displayed good drug loading capacity, we predict that this nanocomposite bioink could also be useful for drug testing or for loading antitumor drugs into scaffolds to replace resected bone tumors.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.msec.2020.110905>.

Abbreviations

Gel gelatin
Alg alginate

nSi nanosilicate
rBMSCs rat bone marrow mesenchymal stem cells
NM normal medium
OIM osteoinductive medium
BLK blank control group without scaffold
0%nSiPPS 3D printed 0%nSi pure Gel-Alg scaffold
0%nSiBPS 3D bioprinted rBMSCs-laden 0%nSi Gel-Alg scaffold
2%nSiPPS 3D printed 2%nSi pure nanocomposite scaffold
2%nSiBPS 3D bioprinted rBMSCs-laden 2%nSi nanocomposite scaffold

CRediT authorship contribution statement

Bin Liu: Conceptualization, Methodology, Investigation, Software, Writing - original draft. **Junqin Li:** Investigation, Writing - review & editing. **Xing Lei:** Investigation, Writing - review & editing. **Pengzhen Cheng:** Resources, Supervision. **Yue Song:** Resources, Supervision. **Yi Gao:** Validation. **Jingzhi Hu:** Resources. **Chunmei Wang:** Resources. **Shuaishuai Zhang:** Resources. **Donglin Li:** Software. **Hao Wu:** Resources. **Hongxun Sang:** Formal analysis. **Long Bi:** Conceptualization, Funding acquisition, Project administration, Supervision. **Guoxian Pei:** Conceptualization, Funding acquisition, Project administration, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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